

# WISHBONE I<sup>2</sup>C Master/ Slave Controller

The WISHBONE I<sup>2</sup>C master/slave controller, because of its simplicity and low cost, is useful for designs that need to communicate with low-speed peripherals.

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## Features

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The WISHBONE I<sup>2</sup>C master/slave controller includes the following features:

- ◆ WISHBONE B.3 interface
- ◆ I<sup>2</sup>C bus speeds of 100 kilobits per second and 400 kilobits per second
- ◆ I<sup>2</sup>C 7-bit addressing
- ◆ Multiple I<sup>2</sup>C masters on one I<sup>2</sup>C bus
- ◆ Up to 256-byte I<sup>2</sup>C transactions
- ◆ Interrupt mode or polling mode
- ◆ Standard WISHBONE interface for the microprocessor
- ◆ Programmable configuration registers

The I<sup>2</sup>C master controller accepts commands from a microprocessor (WISHBONE interface). These commands are decoded into I<sup>2</sup>C slave device read/write cycle transactions. The I<sup>2</sup>C bus transactions can be configured to be 1 to 256 bytes in length. Because the I<sup>2</sup>C master controller can operate in interrupt or polling mode, the microprocessor programmer can poll the I<sup>2</sup>C master for a change in status at periodic intervals or wait to be interrupted by the I<sup>2</sup>C master controller when data needs to be read or written.

## Signals

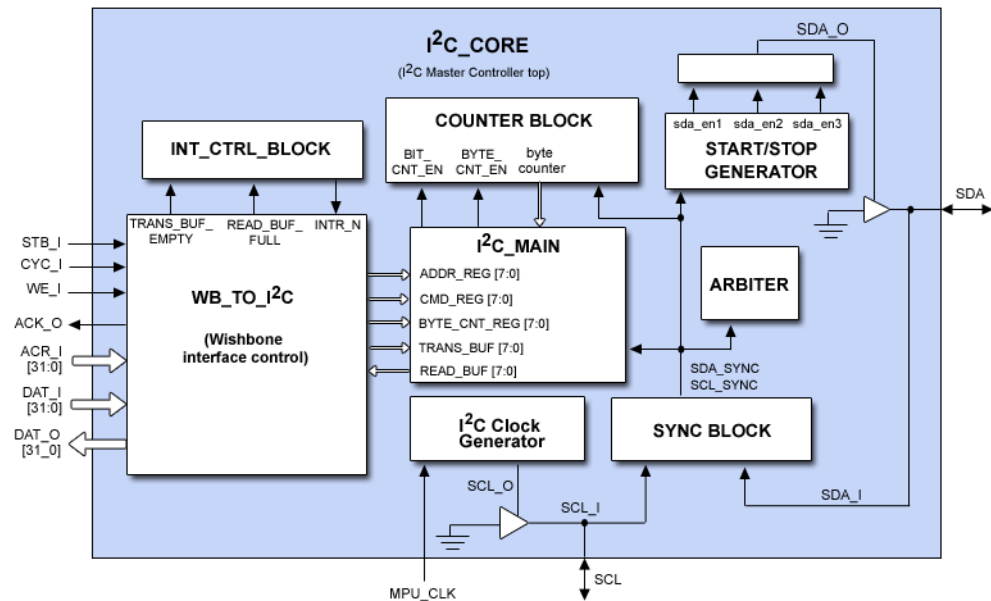
Table 1 describes the signals for the I<sup>2</sup>C controller.

**Table 1: I<sup>2</sup>C Signal Descriptions**

| Signal Name  | Active                | Direction | Description                                                                                                                                                                                   |
|--------------|-----------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CLK_I        | —                     | I         | System clock                                                                                                                                                                                  |
| RST_I        | Low                   | I         | System reset                                                                                                                                                                                  |
| SDA          | —                     | I/O       | Serial data line of the I <sup>2</sup> C bus (tristate)                                                                                                                                       |
| SCL          | —                     | I/O       | Serial clock line of the I <sup>2</sup> C bus (tristate)                                                                                                                                      |
| INTR_N       | Low                   | O         | Interrupt request                                                                                                                                                                             |
| ADR_I[31:0]  | —                     | I         | Address from WISHBONE interface bus                                                                                                                                                           |
| DAT_I [31:0] | —                     | I         | Data input from WISHBONE interface bus                                                                                                                                                        |
| WE_I         | 1 = write<br>0 = read | I         | Write/read signal from WISHBONE interface bus                                                                                                                                                 |
| STB_I        | High                  | I         | Strobe input, indicating that the slave is selected                                                                                                                                           |
| CYC_I        | High                  | I         | Cycle signal, indicating that a valid bus cycle is in progress                                                                                                                                |
| BTE_I        | High                  |           | Burst-type extension signal. The burst-type extension address tag is sent by the master to the slave to provide additional information about the current burst. It is reserved in the design. |
| ACK_O        | High                  | O         | Acknowledge output, indicating termination of a normal bus cycle                                                                                                                              |
| DAT_O [31:0] | —                     | O         | Data output to the WISHBONE interface bus                                                                                                                                                     |

## Module Descriptions

The following sections describe the functions of each of the modules that can be used in the I<sup>2</sup>C controller, shown in Figure 1.

**Figure 1: I<sup>2</sup>C Interface Controller Usage**

## WISHBONE Interface Controller

The WISHBONE interface controller is the main interface between the WISHBONE bus and the I<sup>2</sup>C master controller. It performs the following functions:

- ◆ Accepts write data from the WISHBONE bus and latches the data into the appropriate registers
- ◆ Controls data that is being sent to the WISHBONE bus from the I<sup>2</sup>C master controller
- ◆ Generates the acknowledge signal (**ACK\_O**) of the WISHBONE interface to finish the read or write action
- ◆ Resets the interrupt acknowledge, transmit buffer empty, and read buffer full flags at the appropriate time

## Interrupt Controller

The interrupt controller generates an interrupt signal when either the transmit buffer empty flag or the receiver buffer full flag is set, assuming that the appropriate configuration bits have been set.

## COUNTER\_BLOCK

The counter module contains two counters. The first is a 3-bit counter used for counting each bit of an eight-bit packet transmission or reception during an I<sup>2</sup>C bus packet transaction. The second counter is an eight-bit counter that keeps track of the number of bytes that have been written or read during the I<sup>2</sup>C transaction. It increments after each byte has been written to or read from the I<sup>2</sup>C slave device. The count is then compared with the byte count register. If the values match, the I<sup>2</sup>C master controller considers the transaction complete, issues a stop signal on the I<sup>2</sup>C bus, asserts the done flag, and waits for the next transaction to be initiated from the microprocessor.

## I<sup>2</sup>C Start/Stop Control

The start/stop control block generates and detects start and stop events on the I<sup>2</sup>C bus. The detection of start and stop events is necessary for determining whether the I<sup>2</sup>C bus is in use by another master on the bus when the primary master gets a go signal from the microprocessor. Also, the start detection is necessary for the primary I<sup>2</sup>C master controller because it cannot proceed with a transaction until the start condition has been accepted by the I<sup>2</sup>C bus. As for the start and stop generation block, the I<sup>2</sup>C master controller uses those blocks to generate start and stop events at the appropriate time during the I<sup>2</sup>C transaction.

## I<sup>2</sup>C Clock Generator

This block generates the I<sup>2</sup>C clock signal. You can program the I<sup>2</sup>C master controller to generate an I<sup>2</sup>C clock that will run in either standard mode (100 kilobits per second) or fast mode (400 kilobits per second). The mode depends on the value of the mode bit in the command register. Because of the nature of the I<sup>2</sup>C bus, the actual SCL clock that is seen by all devices on the bus might not be running at the same frequency that the master requests or generates. The I<sup>2</sup>C SCL clock line runs at the speed of the slowest clock generated by a master.

## I<sup>2</sup>C Synchronizer

The synchronizer block synchronizes the SCL\_I and SDA\_I signals with the system clock and protects against metastability. Normally, asynchronous signals can cause a synchronous system to become unstable by violating setup and hold times. To avoid this instability, send the SCL\_I and SDA\_I signals through two registers before using the signals.

## I<sup>2</sup>C Arbiter

The arbiter block contains the code necessary for performing arbitration on the I<sup>2</sup>C bus for designs that use the multi-master feature.

## I<sup>2</sup>C Main State Machine

The main state machine block controls most of the other blocks and serves as the interface between the I<sup>2</sup>C master controller and the I<sup>2</sup>C bus. This block generates many control signals and contains three state machines: main, read, and write. The main state machine, which is where the I<sup>2</sup>C transactions begin, controls the other two state machines. The read and write state machines shift I<sup>2</sup>C data into or out of the appropriate registers. All three state machines are one-hot encoded and are therefore more efficient at driving the various output control signals.

## Register Definitions

The WISHBONE I<sup>2</sup>C controller includes the registers shown in Table 2.

**Table 2: I<sup>2</sup>C Register Definitions**

| Bit                                                          | MPU Direction | Name           | Description                                                                                                                    |
|--------------------------------------------------------------|---------------|----------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Data Buffer</b>                                           |               |                |                                                                                                                                |
| 7-0                                                          | Read          | RX_Data_Buffer | Stores I <sup>2</sup> C receive data                                                                                           |
|                                                              | Write         | TX_Data_Buffer | Stores I <sup>2</sup> C transmit data                                                                                          |
| <b>Address Data Register</b>                                 |               |                |                                                                                                                                |
| 7-1                                                          | Write         | Low_Address_R  | Holds the first seven bits of the I <sup>2</sup> C slave device address                                                        |
| 0                                                            | Write         | I2C_RW_Bit     | I <sup>2</sup> C R/W bit                                                                                                       |
| <b>Command/Status Register (when written to or from MPU)</b> |               |                |                                                                                                                                |
| 7                                                            | Write         | Go             | Starts the I <sup>2</sup> C transaction, which is asserted after the MPU has completed writing to all configuration registers. |
| 6                                                            | Write         | Abort          | Stops a current I <sup>2</sup> C transaction.                                                                                  |
| 5                                                            | —             | Reserved       | Reserved                                                                                                                       |
| 4                                                            | —             | Reserved       | Reserved                                                                                                                       |
| 3                                                            | Write         | I2C_Mode       | 0 – Standard mode, 100 kilobits per second<br>1 – Fast mode, 400 kilobits per second                                           |
| 2                                                            | —             | Reserved       | Reserved                                                                                                                       |
| 1                                                            | Write         | Trans_IE       | Interrupt enable for transmission                                                                                              |
| 0                                                            | Write         | Receive_IE     | Interrupt enable for receive                                                                                                   |
| <b>Command/Status Register (when read from MPU)</b>          |               |                |                                                                                                                                |
| 7                                                            | Read          | I2C_Bus_Busy   | I <sup>2</sup> C bus busy                                                                                                      |
| 6                                                            | Read          | Error          | I <sup>2</sup> C transaction error                                                                                             |
| 5                                                            | Read          | Abort_Ack      | Abort completed by I <sup>2</sup> C controller                                                                                 |
| 4                                                            | Read          | Lost_Arb       | Lost arbitration bit                                                                                                           |

**Table 2: I<sup>2</sup>C Register Definitions**

| Bit               | MPU Direction | Name                | Description                                      |
|-------------------|---------------|---------------------|--------------------------------------------------|
| 3                 | Read          | Done                | Transaction done                                 |
| 2                 | Read          | Reserved            | Reserved                                         |
| 1                 | Read          | Trans_Buffer_Empty  | Transmit buffer empty                            |
| 0                 | Read          | Receive_Buffer_Full | Receive buffer full                              |
| <b>Byte Count</b> |               |                     |                                                  |
| 7-0               | Write         | Byte_Count          | Number of data bytes for the current transaction |
| <b>IAck</b>       |               |                     |                                                  |
| 0                 | Write         | Iack                | Interrupt acknowledge                            |

## Address Map

The microprocessor can control the I<sup>2</sup>C master controller by writing to and reading from configuration registers. The I<sup>2</sup>C master control configuration registers are located at the addresses shown in Table 3.

**Table 3: Configuration Register Addresses**

| A2A1A0 | Description                  |
|--------|------------------------------|
| 000    | Receive/transmit data buffer |
| 001    | Address data register        |
| 010    | Reserved                     |
| 011    | Reserved                     |
| 100    | Command status register      |
| 101    | Byte count                   |
| 110    | Interrupt acknowledge        |